

COMM061: NLP Coursework Report

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Declaration

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1. Overview of the Dataset

The BESSTIE dataset consists of three English varieties: Australian English (en-AU), Indian English (en-IN) and British English (en-UK) [1]. It is designed to support the analysis of sentiment and sarcasm across different varieties of English. Each example is labelled for two binary tasks: Sentiment and Sarcasm. Sentiment is labelled as negative or positive, while Sarcasm is labelled as not-sarcastic or sarcastic. The dataset is already split into train, validation and test sets and these splits are kept fixed in the modelling sections.

1.1 Label Distribution and Dataset Analysis

We analysed the distribution of sentiment and sarcasm labels across the three English varieties using several visualisations. Looking at sentiment first, the dataset is quite balanced overall, with 3,184 negative and 3,059 positive examples. This is also clear in *Figure 1.1* and *Table 1.1*, where all three varieties show very similar proportions. For instance, en-AU has slightly more negative examples (around 54%), en-IN is almost evenly balanced (roughly 51% negative and 49% positive) and en-UK shows a small shift towards positive sentiment (about 52%). Therefore, we can assume that sentiment classification should not be heavily affected by class imbalance.

Sentiment	0	1
variety		
en-AU	0.5401	0.4599
en-IN	0.5073	0.4927
en-UK	0.4845	0.5155

Sarcasm	0	1
variety		
en-AU	0.7058	0.2942
en-IN	0.9318	0.0682
en-UK	0.9237	0.0763

Table 1.1 Proportion of two labels across variety

Sarcasm, however, is clearly imbalanced, with 5,370 (86%) non-sarcastic and only 873 (14%) sarcastic examples. *Figure 1.2* and *Table 1.1* illustrate this, as the proportion of sarcastic examples is much smaller across all varieties. The imbalance is especially strong in en-IN and en-UK, where sarcasm is below 10%, while en-AU gives a sarcasm rate of around 29%. This implies that sarcasm detection will be more challenging, as a model might easily predict "not _sarcastic" most of the time and still achieve high accuracy.



Figure 1.1: Sentiment Distribution by variety

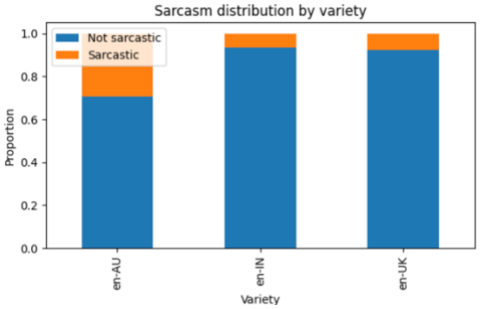


Figure 1.2: Sarcasm Distribution by variety

Figure 1.3 shows how the Sarcasm label is distributed within each Sentiment group. Sarcasm appears much more often in negative-sentiment class across all varieties. For example, in en-AU, around 52% of negative-sentiment texts are sarcastic, while only about 2% of positive texts contain sarcasm. En-IN and en-UK show the same pattern at lower levels with roughly no sarcasm in positive-sentiment examples. This suggests that negative sentiment is a useful cue for Sarcasm, but it is not enough to solve the task since many negative texts are still not sarcastic [2].

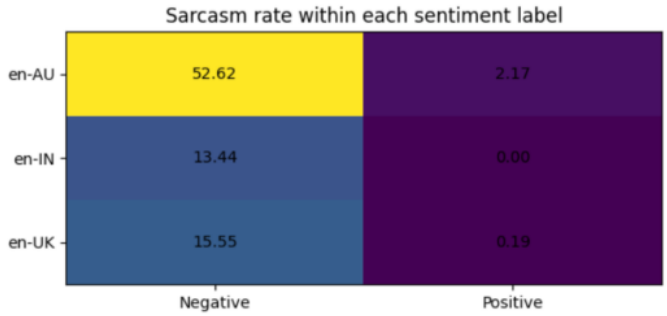


Figure 1.3 Sarcasm rate within each sentiment label across varieties

Another significant factor is the data source. From *Figure 1.4,* Reddit shows much higher sarcasm rates than Google in all varieties. For instance, en-AU has almost 42% sarcasm in Reddit, compared to only 6.70% in Google. Meanwhile, *Figure 1.5* shows that the source mix differs by variety, so differences between en-AU, en-IN and en-UK are not only due to language variation. They may also reflect differences in data source, with Reddit being more informal and sarcasm-heavy than Google review text.

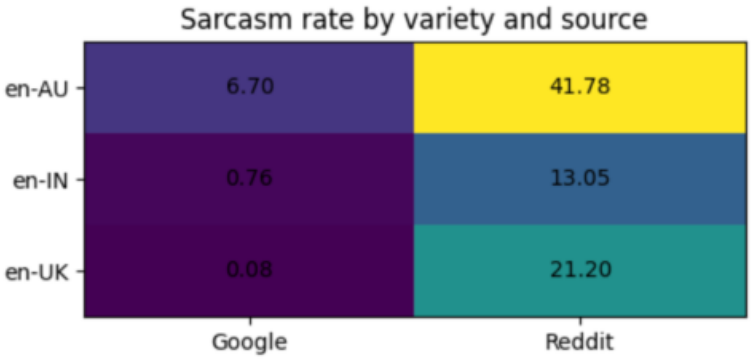


Figure 1.4: Sarcasm rates by variety and data source

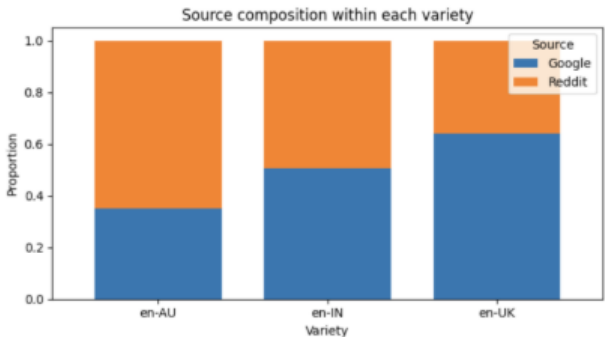


Figure 1.5: Source composition within each variety

We also inspected punctuation that could be linked to sarcasm. According to *Figure 1.6,* ellipses (...) are more common in sarcastic instances for Australian English and particularly UK English, potentially adding an implied tone. Nevertheless, this trend does not hold for Indian English, where non-sarcastic examples use more ellipses. As far as question marks are concerned, in all language varieties, they are more frequent in sarcastic texts, which may point to a questioning style, such as rhetorical or exaggerated questions (*Figure 1.7*). Quotation marks show a similar pattern to question marks, with higher usage in sarcastic texts, particularly in en-AU and en-UK (*Figure 1.8*). Therefore, some punctuation features can act as supporting signals for sarcasm, although the effect is not consistent across all varieties [2].

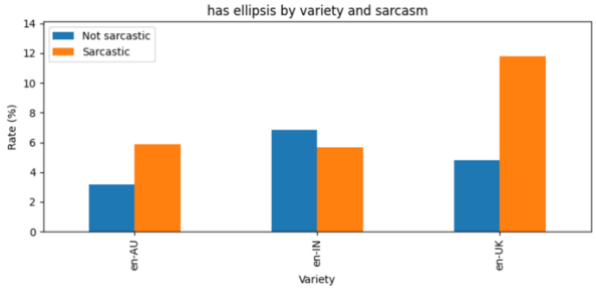


Figure 1.6: Ellipsis usage (%) by sarcasm and variety

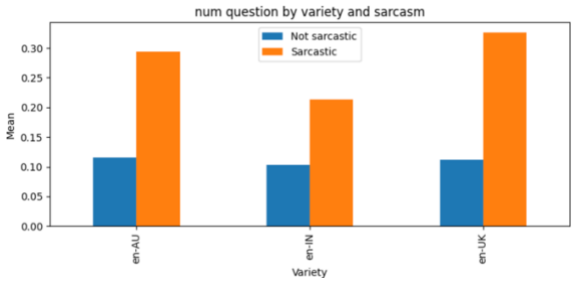


Figure 1.7: Average number of question marks by sarcasm and variety



Figure 1.8: Quotation mark usage (%) by sarcasm and variety

Furthermore, simple linguistic features derived from POS tags and dependency information were analysed using spaCy. As illustrated in *Figures 1.9* and *1.10,* interjection rate is lower overall than negation rate, but it separates sarcastic and non-sarcastic texts more consistently, with higher values for sarcasm across all varieties, especially in en-IN. Negation rate is higher in sarcastic texts in en-AU and slightly higher in en-UK, while there is little difference in en-IN. However, the absolute rates remain low, thus we should treat both features as weak supporting indicators rather than strong signals of sarcasm [2].

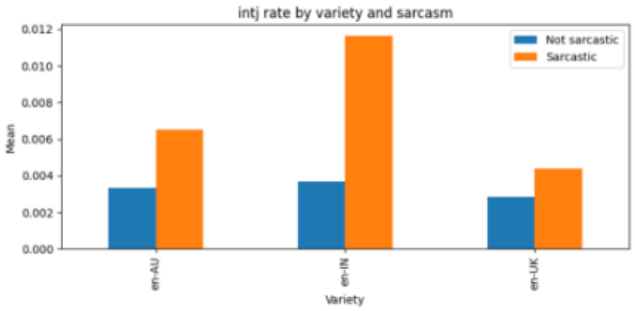


Figure 1.9: Interjection rate by sarcasm across varieties

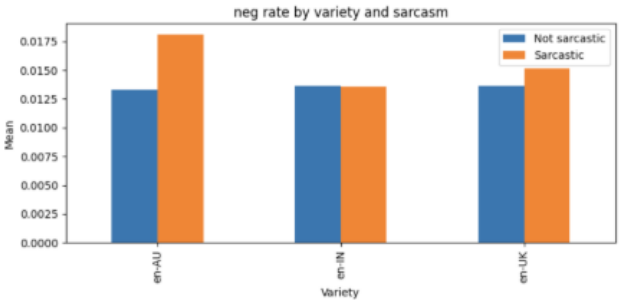


Figure 1.10: Negation rate by sarcasm across varieties

Overall, sentiment appears to be the easier classification task in BESSTIE, as its label is relatively balanced across varieties [1]. In contrast, sarcasm is not explained by only one factor, since class imbalance, source/domain and language variety effects, punctuation and small linguistic features all contribute without giving a simple rule. Thus, the sarcasm label is more context-dependent and harder to identify.

1.2 Vocabulary Analysis and Linguistic Distance

We compare the three varieties using two lexical measures: Jaccard similarity over frequent word types and TF-IDF cosine similarity. Jaccard measures direct vocabulary overlap, while TF-IDF cosine similarity compares the varieties after weighting words by their importance in the corpus [3], [4].

```
Jaccard overlap
UK and IN: 0.3869
AU and IN: 0.3771
UK and AU: 0.4484
```

	en-UK	en-AU	en-IN
en-UK	1.0000	0.4484	0.3869
en-AU	0.4484	1.0000	0.3771
en-IN	0.3869	0.3771	1.0000

Table 1.2: Jaccard similarity between English varieties

Using Jaccard similarity (*Table 1.2*), en-UK and en-AU show the highest overlap (0.4484), while pairs involving en-IN are lower (0.3869 for en-UK/en-IN and 0.3771 for en-AU/en-IN). This indicates that Indian English shares less vocabulary with the other two varieties, whereas UK and Australian English are lexically closer.

We can observe the same pattern in TF-IDF cosine similarity results (*Figure 1.11*). More specifically, en-UK and en-AU are again the closest pair, with a similarity of 0.911, while the pairs involving en-IN are lower: 0.824 for en-UK/en-IN and 0.7809 for en-AU/en-IN. Since TF-IDF gives

more weight to words that are more informative, this suggests that en-IN differs not only in raw vocabulary overlap, but also in the words that carry more weight in the dataset [3].

The distinctive-token analysis helps explain the lexical distance involving en-IN, which includes tokens such as hai, ka, rs, dosa, biryani and masala. These tokens reflect local and cultural vocabulary and show that the lower overlap is driven by meaningful lexical variation rather than simply random or rare words.

When we examine Google and Reddit separately (*Table 1.3),* the variety pattern changes. In the Google data, en-UK and en-AU remain closest, especially with TF-IDF cosine similarity. In the Reddit data, the similarity scores are closer together, and the Jaccard pattern is less stable, thus the measured lexical distance is also shaped by source and topic differences. Reddit adds more informal and varied vocabulary, which makes the variety pattern less clear than in Google reviews.

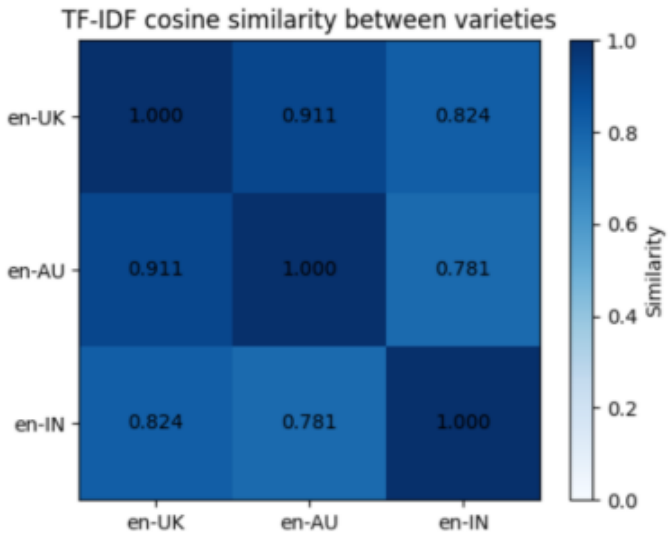


Figure 1.11: TF-IDF cosine similarity between varieties

Google
Jaccard
UK and AU: 0.4461
UK and IN: 0.374
AU and IN: 0.3868
TF-IDF cosine similarity

	en-UK	en-AU	en-IN
en-UK	1.0000	0.9616	0.7963
en-AU	0.9616	1.0000	0.7995
en-IN	0.7963	0.7995	1.0000

Reddit
Jaccard
UK and AU: 0.3246
UK and IN: 0.3366
AU and IN: 0.3194
TF-IDF cosine similarity

	en-UK	en-AU	en-IN
en-UK	1.0000	0.8407	0.8205
en-AU	0.8407	1.0000	0.8230
en-IN	0.8205	0.8230	1.0000

Table 1.3: Jaccard and TF-IDF similarity scores by variety and data source

Linguistic distance refers to how different language varieties are in terms of vocabulary, structure and usage. In our analysis, we focus on lexical distance, using vocabulary overlap and TF-IDF cosine similarity. The results show that en-IN is more lexically distant from en-UK and en-AU, while en-UK and en-AU are closer to each other. The distinctive tokens and the Google/Reddit split suggest that this distance is influenced by local vocabulary and data source, not only by English variety. These differences can still have an impact on model performance, as models may rely on vocabulary patterns that do not transfer well across varieties or sources.

2. Experiments

2.1 Baseline/PTLM Gap

For the baseline model, we used TF-IDF features with Logistic Regression. Before selecting the final baseline, we compared Logistic Regression with Linear SVM on the validation set. Both models were tested with C values from 0.001 to 8.0, and validation scores were averaged across three seeds. We also compared word-level TF-IDF with a hybrid setup that combined word n-grams and character n-grams. The word-level features used unigrams and bigrams, while the hybrid setup added character n-grams to capture smaller spelling and stylistic patterns.

Linear SVM achieved the strongest Sentiment validation result, reaching Macro-F1 0.8690 with hybrid TF-IDF features. However, Logistic Regression performed better on Sarcasm, reaching Macro-F1 0.6647 with word-level TF-IDF and balanced class weights. As Sarcasm is the more challenging and imbalanced task, and the difference in Sentiment performance was small, Logistic Regression was selected as the final baseline model for both tasks.

For the final Logistic Regression configuration, Sentiment performed best with word-level TF-IDF, no class weighting and C=8.0. For Sarcasm, the best configuration used word-level TF-IDF, balanced class weights and C=0.5. This matches the label distribution in the dataset, since Sentiment is relatively balanced while Sarcasm needs class weighting because the sarcastic label is much smaller. The final baseline models were trained on the combined training and validation data before evaluation on the fixed test set.

For the PTLM comparison, we fine-tuned RoBERTa-base on the same dataset splits. RoBERTa-base was used to provide a direct contrast with TF-IDF features. The baseline mainly represents surface word patterns, while RoBERTa-base uses contextual representations, where the meaning of a word can change depending on the surrounding text. Inputs were tokenised to a maximum length of 128 and RoBERTa-base was fine-tuned with batch size 8. We tuned the learning rate and number of epochs on the validation set. For Sarcasm, we also tested whether to use class-weighted loss. The selected Q2.1 RoBERTa settings were learning rate 3e-5 and 2 epochs for Sentiment, and learning rate 2e-5 and 3 epochs for Sarcasm; both selected settings used the standard unweighted loss. After tuning, the final RoBERTa-base runs were trained on the combined training and validation data before evaluation on the fixed test set.

Based on *Table 2.1* and *Table 2.2*, RoBERTa-base performs better on Sentiment, achieving Macro-F1 0.8908 compared with 0.8435 for the baseline. This suggests that the PTLM is useful when the label is more directly expressed in the wording. However, the pattern is different for Sarcasm. The baseline slightly outperforms RoBERTa-base, with Macro-F1 scores of 0.6251 and 0.6075 respectively. Thus, the pre-trained model helps clearly for Sentiment, but not for Sarcasm in this setup.

Overall test summary

	macro_precision		macro_recall		macro_f1	
	mean	std	mean	std	mean	std
task						
Sarcasm	0.6170	0.0002	0.7028	0.0006	0.6251	0.0001
Sentiment	0.8446	0.0000	0.8432	0.0000	0.8435	0.0000

Table 2.1. TF-IDF + LR test performance for both tasks

	macro_precision		macro_recall		macro_f1	
	mean	std	mean	std	mean	std
task						
Sarcasm	0.6822	0.0275	0.5920	0.0524	0.6075	0.0565
Sentiment	0.8909	0.0032	0.8908	0.0029	0.8908	0.0030

Table 2.2. RoBERTa-base test performance for both tasks

A likely reason is that Sarcasm is less directly tied to individual words [2]. A sarcastic example can contain positive, neutral or exaggerated wording while meaning something different from the literal text. It is also affected by class imbalance and variety-specific usage [2]. Hence, RoBERTa-base gives

a clear gain for Sentiment, while the simpler TF-IDF baseline remains competitive for Sarcasm, especially because its class weighting makes it more sensitive to the minority label.

2.2 Cross-Variety Evaluation

This experiment investigates whether a sarcasm detection model trained on one variety of English can generalize effectively to other varieties. We use RoBERTa-base for the sarcasm classification task, as it's a widely used pretrained encoder model that performs well on sequence classification problems [5]. The model architecture and task setup were kept consistent across all runs, thereby any differences in the performance mainly reflect the influence of the training variety and the characteristics of the data within that variety.

RoBERTa-base was fine-tuned separately on each of the three varieties: AU, UK and IN. For each case, the model was trained using the training split of the selected variety and validated on the corresponding validation split. After training, each model was evaluated on all three test sets. This produced a full 3x3 train/test matrix which includes both in-variety results and cross variety results.

The same training config was used across all runs to ensure consistency. A learning rate of 2e-5 was used and trained for 3 epochs, following the selected RoBERTa setup for sarcasm. In addition, weighted loss was enabled to reduce the impact of class imbalance, as the dataset contains a much larger proportion of non-sarcastic examples [6]. Inputs were tokenized with a maximum sequence length of 128 and trained using a batch size of 8. Validation was performed during training with early stopping applied to prevent overfitting. Each experiment was repeated across the random seeds, and the results are reported using the mean Macro F1 score.

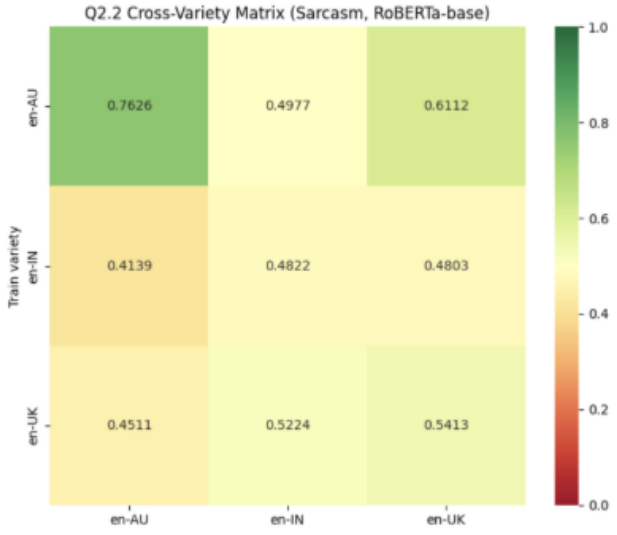


Figure 2.1: Cross-variety Macro-F1 scores for sarcasm using RoBERTa-base.

The main goal of this setup is to evaluate transferability across varieties. If sarcasm patterns were similar across different forms of English, we would expect a model trained on one variety to perform well on others. However, it's seen that this isn't the case. The strongest performance is observed in the en-AU to en-AU setting, where the model achieves a Macro F1 score of 0.7626, as shown in *Figure 2.1*. More broadly, the average in-variety performance is 0.5954, while the cross variety average drops to 0.4961. This indicates that models generally perform better when the training and test data come from the same variety and that performance decreases when applied to unseen varieties.

The pattern is not as simple as saying that AU and UK always transfer better together. The en-AU-trained model performs better on en-UK than on en-IN, with Macro-F1 0.6112 compared with 0.4977, which is consistent with the lexical similarity pattern in *Section 1.2*. However, the en-UK-trained model

performs better on en-IN than on en-AU, with Macro-F1 0.5224 compared with 0.4511. This means lexical similarity helps explain part of the result, but not all of it. Class distribution, data source and the examples available within each variety also affect transfer.

The en-IN-trained model is the clearest weak point. It performs poorly across all three test varieties and collapses towards the majority non-sarcastic label. This shows that the issue is not only whether a model transfers across varieties, but also whether the training variety gives the model enough useful signal to learn the sarcastic label in the first place.

Several factors may explain these results. Differences in language use across varieties mean that sarcasm can be expressed through different vocabulary and stylistic patterns, making transfer more difficult [6]. In addition, class imbalance plays an important role, as the scarcity of sarcastic examples leads the model to favour the majority non-sarcastic class, reducing recall for sarcasm in cross-variety settings [6].

Overall, these findings suggest that cross-variety performance is influenced by a combination of linguistic differences, data characteristics, and label imbalance, rather than model capacity alone [7]. Improving generalisation would likely require more balanced data, better representation of variety-specific features, or training strategies that explicitly account for variation between English varieties [6].

2.3 Parameter-Efficient Fine-Tuning of TinyLlama-1.1B

Since most weights encode broad linguistic knowledge, fully fine-tuning 1B+ parameter Large Language Models (LLMs) for particular dialectal types is computationally expensive and wasteful in terms of architecture [8]. In this work, the underlying model is kept frozen while trainable low-rank matrices are injected into specific attention layers using Low-Rank Adaptation (LoRA).
Base Model: To meet the 1-3B parameter constraint, TinyLlama/TinyLlama-1.1B-Chat-v1.0 was chosen [9].
Variety Specialization: Three distinct adapters (en-AU, en-IN, and en-UK) were trained instead of averaging patterns over a pooled dataset. This enables each adaptation to focus on the distinct lexical and pragmatic sarcasm indicators of its particular variety.

2.3.1 Experimental Setup and Parameter Efficiency

The query q_proj and value v_proj projection matrices were the focus of the LoRA configuration since they are the most variety-sensitive parts as they determine how the model responds to context.

Parameter	Value	Details
Rank (r)	8	Balance between expressiveness and parameter count.
Scaling	16	Controls adapter influence.
Trainable Params	1,130,496	0.11% of total weights (1.03B).
Frozen Params	1,034,516,480	99.89% of weights remained unchanged.

Table 2.3: Lora Parameter Values

To minimise training time, training data was limited to 500 samples per variety. To address significant class imbalance, weighted cross-entropy was used, especially in en-IN (6.8% sarcastic) and en-UK (7.6% sarcastic). The sarcastic class received a weight of 6.579 for the en-IN variety, whereas the majority class received a weight of 0.541. Multi-Seed validation (Macro-F1) and selection are displayed in *Table 2.4*.

Variety	Seed 42	Seed 0	Chosen Configuration
en-AU	0.7022	0.7321	Seed 0
en-IN	0.6866	0.6866	Seed 42 (Tie)
en-UK	0.6844	0.7110	Seed 0

Table 2.4 Multi-Seed validation Macro-F1

2.3.2 Results and Cross-Adapter Analysis

The performance among dialects is shown in the 3x3 Macro-F1 matrix. The diagonal entries underline the dialect transfer gap and indicate in-variety performance. The heatmap graphically validates the Indian adapter's disastrous failure and diagonal dominance on en-AU.

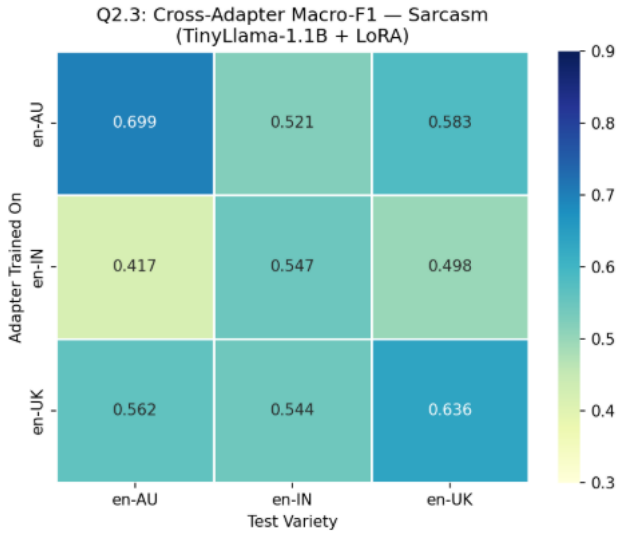


Figure 2.1 3×3 Cross Variety Macro-F1 matrix for Sarcasm Detection

en-AU Adapter exhibits the strongest in-variety performance (0.699), benefiting from the highest sarcasm prevalence (29.4%). It transfers significantly better to en-UK than en-IN, suggesting shared inner-circle sarcasm patterns such as irony and understatement.

en-IN Adapter shows the weakest in-variety result (0.547) and the poorest cross-variety transfer (0.417 on AU). The extreme class imbalance during training creates a bias toward not sarcastic predictions that collapses performance when applied to varieties with higher sarcasm density.

en-UK Adapter achieves a balanced in-variety score of 0.636. Consistent with the en-AU adapter, it transfers more effectively to the Australian variety than the Indian variety, reinforcing the inner-circle similarity hypothesis.

Variety-specific training offers a quantifiable empirical benefit because the diagonal outcomes are consistently the highest in each row. However, en-IN consistently performs poorly on all test sets, indicating that data scarcity rather than model architecture remains the key constraint.

2.3.3 Comparison with Prior Baselines

LoRA variety-specific adapters were compared against pooled TF-IDF and RoBERTa-base models and Macro-F1 scores are shown in the table 2.3.

Setup	en-AU	en-IN	en-UK	Overall Mean
Q2.1 TF-IDF + LogReg (Pooled)	0.682	0.550	0.578	0.6251
Q2.2 RoBERTa-base (Pooled)	0.590	0.523	0.634	0.6075
Q2.3 TinyLlama + LoRA (Specific)	0.699	0.547	0.636	0.6275

Table 2.3 Comparative Sarcasm detection performance using Macro-F1 across models

1. LoRA uses only 0.11% of trainable parameters, matching the TF-IDF baseline overall (0.6275 vs. 0.6251).
2. On en-AU, the LoRA method greatly beat pooled RoBERTa, showing that specialization is beneficial when there is enough signal.
3. En-IN continues to be a consistent failure point (0.547–0.550) for all architectures, indicating that this is a data scarcity issue rather than a model restriction.

2.3.4 Deployment and Scaling Advantages

The LoRA technique has important deployment-related engineering benefits:

1. Storage Efficiency: A whole model checkpoint takes about 2.2 GB, whereas each variety-specific adaptor takes about 4 MB.
2. Inference Latency: All three adapters can be loaded into a single model instance using the PEFT library. Variety switching has almost no delay since moving across dialects just requires a pointer swap rather than a model reload.
3. Feasibility: High-performance LLM adaptation is possible without large compute budgets, as shown by the fact that each adapter can be trained in a matter of minutes.

3. Evaluation

We evaluated the final test results from the three experiment setups using macro-F1 as the main metric, as the sarcasm task is imbalanced and accuracy values would not show whether the model is actually detecting the minority sarcastic class. Nevertheless, we consider precision, recall, per-class F1 and confusion matrices as well, to better understand model behaviour and the types of errors made.

3.1 Baseline and RoBERTa evaluation

Firstly, *Table 3.1* compares the TF-IDF + Logistic Regression baseline with fine-tuned RoBERTa-base. On Sentiment, both models perform strongly, but RoBERTa increases Macro-F1 from 0.8435 to 0.8908, showing that in this task, the transformer model beats the baseline. However, that is not true for Sarcasm, as there is a performance drop for both models, with the baseline reaching a Macro-F1 of 0.6251 and RoBERTa reaching 0.6075. Thus, RoBERTa improves Sentiment performance, but not Sarcasm Macro-F1.

	task	model	macro_precision	macro_recall	macro_f1	macro_f1_std
0	Sentiment	baseline	0.8446	0.8432	0.8435	0.0000
1	Sarcasm	baseline	0.6170	0.7028	0.6251	0.0001
2	Sentiment	roberta	0.8909	0.8908	0.8908	0.0030
3	Sarcasm	roberta	0.6822	0.5920	0.6075	0.0565

Table 3.1: Q2.1 summary metrics for TF-IDF + Logistic Regression and RoBERTa-base

Furthermore, *Table 3.2* reports the per-label results for the two tasks. Baseline's F1 for the non-sarcastic label is 0.8370, whereas for the sarcastic one, it is only 0.4131. RoBERTa non-sarcastic F1 is higher at 0.9214, while sarcastic F1 score is slightly larger than the baseline (0.4215). Hence, both models are much better at recognising non-sarcastic instances. This could be due to class imbalance, making it harder for the model to correctly identify sarcasm.

The main trade-off is precision versus recall for the sarcastic class. The baseline has lower sarcastic precision (0.3043), but higher sarcastic recall (0.6426). RoBERTa has higher sarcastic precision, (0.5069), but much lower sarcastic recall (0.3607). Therefore, the baseline catches more sarcastic examples, but also incorrectly predicts sarcasm more often. RoBERTa makes fewer false positive sarcasm predictions, but it misses more true sarcastic examples.

	task	model	class	precision	recall	f1	support
0	Sentiment	baseline	negative	0.8328	0.8693	0.8506	1117
1	Sentiment	baseline	positive	0.8564	0.8171	0.8363	1066
2	Sarcasm	baseline	not_sarcastic	0.9292	0.7614	0.8370	1878
3	Sarcasm	baseline	sarcastic	0.3043	0.6426	0.4131	305
4	Sentiment	roberta	negative	0.8863	0.9069	0.8965	1117
5	Sentiment	roberta	positive	0.9000	0.8780	0.8889	1066
6	Sarcasm	roberta	not_sarcastic	0.9008	0.9430	0.9214	1878
7	Sarcasm	roberta	sarcastic	0.5069	0.3607	0.4215	305

Table 3.2: Q2.1 per-class precision, recall and F1

This can be also observed through *Figure 3.1*. In the Sarcasm confusion matrix, the baseline correctly predicts 196 sarcastic examples and misses 109. RoBERTa correctly predicts only 110 sarcastic examples and misses 195. Therefore, a significant limitation of RoBERTa is its low recall for the sarcastic class.

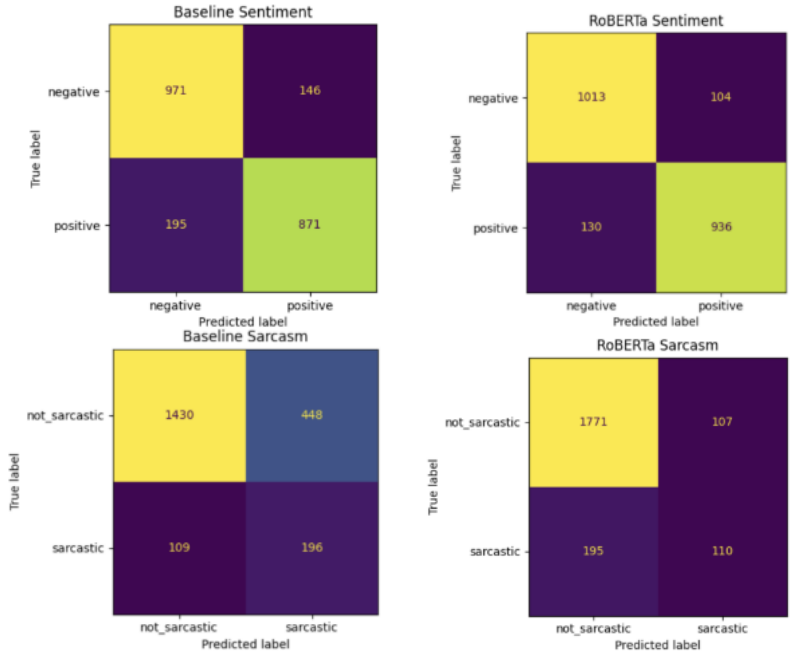


Figure 3.1: Confusion matrices for the Q2.1 TF-IDF + Logistic Regression baseline and RoBERTa-base models on Sentiment and Sarcasm.

3.2 Cross-variety RoBERTa evaluation

The cross-variety experiment fine-tunes one RoBERTa-base separately on en-AU, en-IN and en-UK. Each trained model is then evaluated on all three test varieties. *Figure 3.2* shows the Macro-F1 results for all train-test combinations on the sarcasm task. The en-AU in-variety experiment gives the best performance, where the Macro-F1 is 0.7626. In contrast, the model trained on en-IN shows weak performance across all test varieties, including en-IN itself. The in-variety average is 0.5954, while the cross-variety average is 0.4961 (*Table 3.3*), demonstrating that performance generally drops when the training and test varieties differ. Therefore, the model struggles to generalise sarcasm across English varieties.

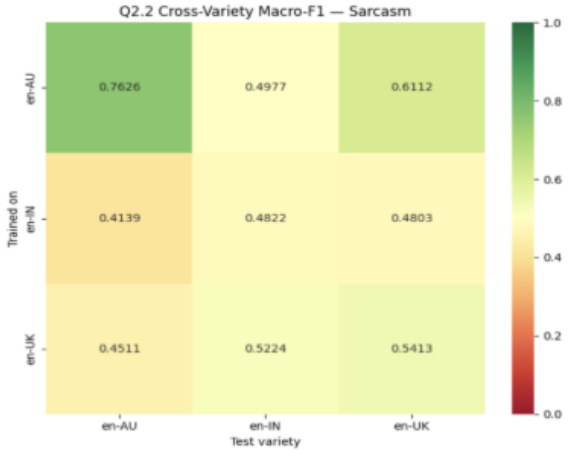


Figure 3.2.Cross-variety RoBERTa-base Macro-F1 for Sarcasm. Rows show training variety; columns test variety

	same_f1	cross_f1	drop
0	0.5954	0.4961	0.0993

Table 3.3: Same-variety vs cross-variety Macro-F1 and performance drop for sarcasm

Moreover, the cross-variety pattern is not consistent across train-test pairs based on *Table 3.4*. The en-AU-trained model scores higher on en-UK(0.6112) than on en-IN(0.4977), which could suggest that Australian and British English are closer for this setup than Australian and Indian English. This may be due to similarities in expressions, slang, and language use. Interestingly, the en-UK-trained model

shows a different pattern, performing better on en-IN than on en-AU, with Macro-F1 of 0.5224 and 0.4511, respectively. This indicates that transfer is not always consistent and does not follow a simple pattern.

	train	test	macro_precision	macro_recall	macro_f1	macro_f1_std
0	en-AU	en-AU	0.7532	0.7829	0.7626	0.0120
1	en-AU	en-IN	0.5527	0.6964	0.4977	0.0087
2	en-AU	en-UK	0.6122	0.8410	0.6112	0.0033
3	en-IN	en-AU	0.3531	0.5000	0.4139	0.0000
4	en-IN	en-IN	0.4657	0.5000	0.4822	0.0000
5	en-IN	en-UK	0.4621	0.5000	0.4803	0.0000
6	en-UK	en-AU	0.4481	0.5157	0.4511	0.0644
7	en-UK	en-IN	0.5054	0.5563	0.5224	0.0696
8	en-UK	en-UK	0.5515	0.5436	0.5413	0.1056

Table 3.4: cross-variety macro-precision, recall, and F1 scores for RoBERTa across all train-test variety combinations.

Table 3.5 presents the per-label results for the cross-variety RoBERTa experiment. The clearest failure is the model trained on en-IN. Its F1 and recall for the sarcastic label are both 0.0000 across all three test varieties, which means it fails to identify any sarcastic examples and is effectively predicting only the majority class. The model trained on en-UK is also weak on sarcastic recall, but it does not collapse as completely as the en-IN-trained model, as it still identifies a small number of sarcastic examples.

	train	test	macro_f1	not_f1	not_r	sar_f1	sar_r	sar_miss
0	en-AU	en-AU	0.7626	0.8434	0.8004	0.6818	0.7653	0.2347
1	en-AU	en-IN	0.4977	0.7696	0.6368	0.2258	0.7559	0.2441
2	en-AU	en-UK	0.6112	0.8513	0.7450	0.3710	0.9371	0.0629
3	en-IN	en-AU	0.4139	0.8278	1.0000	0.0000	0.0000	1.0000
4	en-IN	en-IN	0.4822	0.9645	1.0000	0.0000	0.0000	1.0000
5	en-IN	en-UK	0.4803	0.9607	1.0000	0.0000	0.0000	1.0000
6	en-UK	en-AU	0.4511	0.8266	0.9837	0.0756	0.0476	0.9524
7	en-UK	en-IN	0.5224	0.9474	0.9579	0.0974	0.1548	0.8452
8	en-UK	en-UK	0.5413	0.9606	0.9928	0.1220	0.0943	0.9057

Table 3.5: class-level Sarcasm results across train-test varieties.

Figure 3.3 shows two different error patterns from the RoBERTa variety experiment. The en-AU in-variety run detects both labels reasonably well, correctly identifying 154 sarcastic examples while missing 42. In contrast, the model trained on en-IN and evaluated on en-AU predicts every test example as non-sarcastic. It correctly classifies all 471 non-sarcastic examples but fails to detect any of the 196 sarcastic examples, which indicates that in this setting, the model completely fails to identify sarcasm.

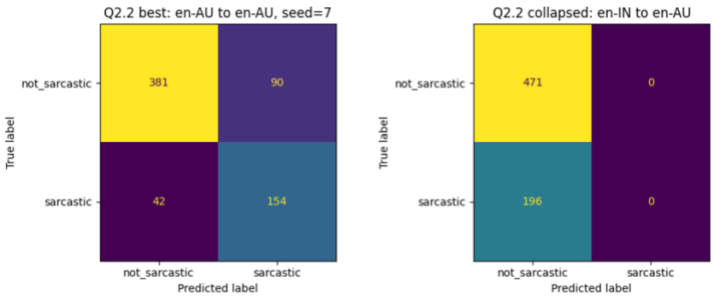


Figure 3.3. Representative Q2.2 confusion matrices. The left matrix shows the best en-AU to en-AU run. The right matrix shows the collapsed en-IN to en-AU run.

3.3 LoRA adapter evaluation

We then evaluate TinyLlama-1.1B with LoRA adapters trained separately for each variety. *Figure 3.4* shows the cross-adapter Macro-F1 matrix, where each adapter is tested on all three varieties. The best LoRA result is again en-AU to en-AU, with Macro-F1 0.6989. The en-UK in-variety adapter achieves relatively good performance as well, with Macro-F1 0.6362, whereas the en-IN one gives a lower Macro-F1 value of 0.5473.

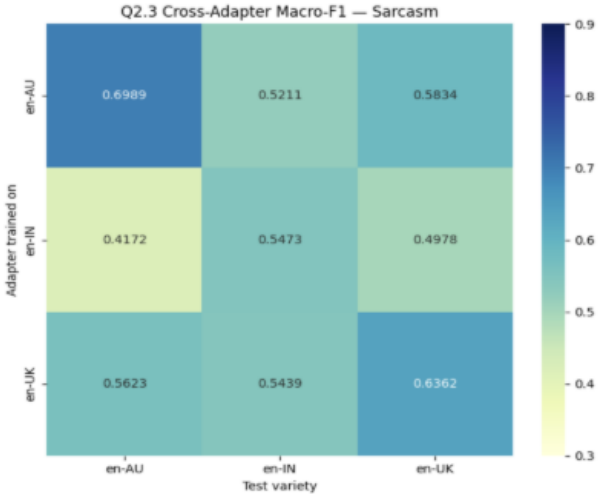


Figure 3.4. Cross-adapter Macro-F1 for TinyLlama-1.1B + LoRA on sarcasm. Rows: train variety; columns: test variety

The in-variety LoRA average is 0.6275, while the cross-variety average is 0.5210, giving a drop of 0.1065. Thus, there is a performance decline when the adapters are used across varieties. The per-label LoRA results (Table 3.6) show that the adapters still struggle with sarcastic examples. For en-AU to en-AU, the sarcastic F1 is 0.600 and sarcastic recall is 0.684, which is the strongest in-variety LoRA result. For en-UK to en-UK, sarcastic F1 falls to 0.347, with recall 0.472. The en-IN to en-IN adapter performs worst on the sarcastic class, with sarcastic F1 0.129 and recall 0.071. This means the en-IN adapter mostly predicts the majority non-sarcastic class.

	train	test	macro_f1	not_f1	not_r	sar_f1	sar_r	sar_miss
0	en-AU	en-AU	0.6989	0.798	0.752	0.600	0.684	0.316
1	en-AU	en-IN	0.5211	0.839	0.749	0.204	0.500	0.500
2	en-AU	en-UK	0.5834	0.842	0.739	0.325	0.811	0.189
3	en-IN	en-AU	0.4172	0.824	0.992	0.010	0.005	0.995
4	en-IN	en-IN	0.5473	0.966	0.997	0.129	0.071	0.929
5	en-IN	en-UK	0.4978	0.960	0.997	0.036	0.019	0.981
6	en-UK	en-AU	0.5623	0.801	0.877	0.323	0.250	0.750
7	en-UK	en-IN	0.5439	0.916	0.892	0.172	0.232	0.768
8	en-UK	en-UK	0.6362	0.925	0.898	0.347	0.472	0.528

Table 3.6: Per-label Sarcasm results for the in-variety+ TinyLlama-1.1B+LoRA adapters

This can be seen in Figure 3.5 as well. The en-AU adapter detects134 sarcastic examples and misses 62, while the en-UK one detects 25 and misses 28 and the en-IN one identifies only 4 and misses 52.

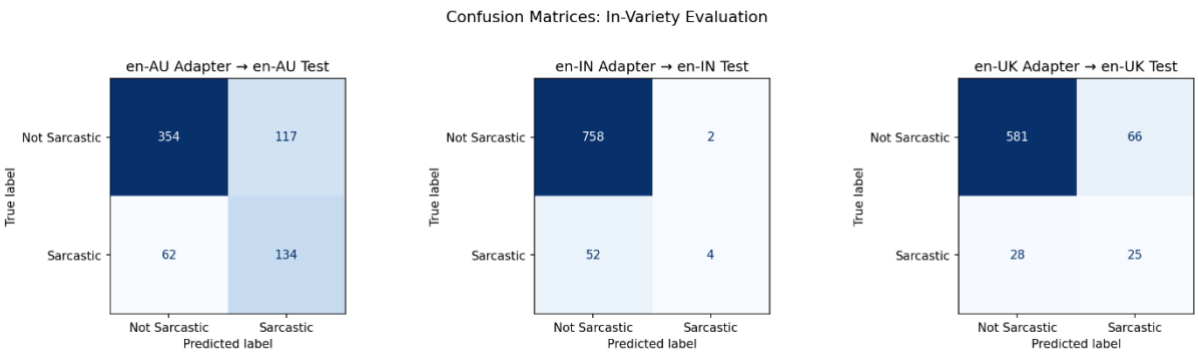


Figure 3.5: Confusion matrices for in-variety LoRA adapters

Figure 3.6 also shows the large gap between non-sarcastic F1 and sarcastic F1 for all the in-variety adapters. More specifically, this gap is smallest for en-AU and largest for en-IN.

In conclusion, LoRA gives useful in-variety performance, especially for en-AU, but it does not solve the minority class issue. The en-IN adapter remains the weakest case.

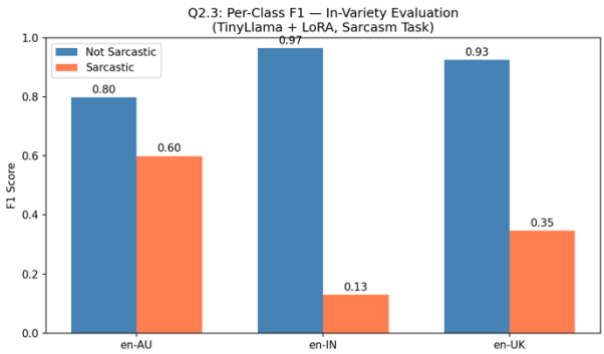


Figure 3.6: Per-class F1 for in-variety LoRA adapters

3.4 Overall comparison

	setup	model	macro_f1	sar_f1	sar_recall	sar_miss
0	Q2.1 Baseline (TF-IDF+LR)	TF-IDF+LR	0.6251	0.4131	0.6426	0.3574
1	Q2.1 RoBERTa (best run)	RoBERTa	0.6714	0.4215	0.3607	0.6393
2	Q2.2 Cross-variety (best mean)	en-AU to en-AU	0.7626	0.6818	0.7653	0.2347
3	Q2.3 LoRA (best adapter-test)	en-AU to en-AU	0.6989	0.6000	0.6840	0.3160

Table 3.7: Final Sarcasm comparison across Q2.1, Q2.2 and Q2.3.

Based on *Table 3.7,* which summarises the main results from the earlier evaluation figures and tables, there is not one model that is best in every sense. If we only consider the highest single Macro-F1 score, the strongest result is the RoBERTa model trained and tested on en-AU, with Macro-F1 0.7626, as shown in *Figure 3.*2 and *Table 3.4.* This shows that RoBERTa can work well when the training and test variety match, especially for en-AU.

However, this does not mean RoBERTa is always the best choice. In the pooled Sarcasm comparison in *Table 3.1,* the TF-IDF + Logistic Regression baseline slightly outperforms RoBERTa-base, with Macro-F1 0.6251 compared with 0.6075. *Table 3.2* and *Figure 3.1* show that the baseline also has higher recall for the sarcastic label, meaning it catches more true sarcastic examples, although it also produces more false positives. This matters, as the aim is not only to improve the average score, but also to identify the minority sarcastic class.

The best LoRA setting is also en-AU in-variety, with Macro-F1 0.6989, as shown in *Figure 3.4*. This is lower than the best RoBERTa en-AU result, but it is higher than pooled RoBERTa and close to the TF-IDF baseline overall. The matched LoRA average is 0.6275, as summarised in *Table 3.7*, which is only slightly above the baseline's 0.6251. Thus, LoRA provides a useful variety-specific approach, but it does not clearly dominate the other models.

Overall, the results are mixed but still informative. RoBERTa gives the strongest single score in the en-AU matched setting, LoRA works well when the adapter matches the variety and the TF-IDF baseline remains competitive, particularly when the priority is detecting sarcasm. The main limitation is that performance varies significantly across varieties, and weaker settings tend to miss the sarcastic class or predict the majority non-sarcastic label too often. Therefore, the final model choice depends on the aim: highest Macro-F1, best variety-specific adapter, or stronger recall for sarcasm.

4. Sarcasm Explanation and Error Analysis

4.1 Misclassified Examples

Text No.	Text	True Label	Predicted Label
1	How insulated are they? I'm over paper thin walls.	0	1
2	That's ridiculous. But because people are too embarrassed/shy to say not thanks, it sells at that price and just encourages all prices to be increased. Don't be afraid to walk away if you don't like the price.	0	1
3	Bought an ANGUS Bacon BBQ Sauce with Onion and cheese. DRY MEAT!, NO ONION! CHEESE yes, 1 BACON Not 2, DOUBLE BBQ SAUCE! My friend has NOT had McDonald for years, and Now WILL NOT eat it again.	0	1
4	Yes, if nuclear power plants were free and everyone was happy to have one nearby, they'd threaten all coal power, even some wind and solar. But they're not and they aren't. It's nothing to do with not being "allowed". Literally no energy company is asking for "permission", because they know that nuclear makes no business sense.	0	1
5	Yes, and wear ugly shoes, lease a BMW and be hated by your own mum	1	0
6	Yes, "resignation" should not be required - just mark the person as "do not roster". Even if you did have to rehire to cover them, for casuals it doesn't matter, because there is no requirement to take them back after their holiday - that would be a simple decision when they came back of "do I need staff now, or not?" Only reason I can think of is that it would be paperwork to explain why the person was marked "no longer employed" in the payroll and/or rostering computer system. Keeps it neat and tidy?	1	0
7	Department of Honesty aka DoH!	1	0
8	so you're a genius and a mind reader WOW.	1	0
9	How do you define a "cold country"? We are talking San Francisco weather.	1	0
10	Tram way back in the 90s, hit a car. After school and packed. I was holding on, my friend was not. I saw him yeet, down the aisle as if he were falling off a cliff, taking the whole aisle with him. He maintained eye contact the whole way, wide eyed with an outstretched hand. One of those slow motion moments. A++ would experience again.	1	0

Table 4.1 Ten misclassified sarcasm examples produced by the TinyLlama LoRA adapter trained on the en-AU variety

4.2 Selection and Analysis of Chosen Examples for Few-Shot Prompting

From the ten misclassified instances in Table 4.1, four examples were selected to analyse in detail and construct a few-shot prompt. These examples were chosen because they clearly illustrate different types of model confusion, including implicit sarcasm, exaggerated tone, and conversational ambiguity. From Table 4.1, four examples were selected (Texts 3, 5, 9, and 10) to construct a few-shot prompt.

Text 3 is labelled as not sarcastic (0). This example represents a genuine complaint expressed with strong emphasis through capitalisation, repetition, and exaggerated frustration. Despite its dramatic tone, the statement is intended literally rather than sarcastically, as the speaker is clearly dissatisfied with the product. The model incorrectly predicts sarcasm in this case, likely because it over-associates intense emotional language and exaggeration with sarcastic intent, resulting in a false positive.

Text 5 is labelled as sarcastic (1). This statement conveys sarcasm through exaggerated and unrealistic consequences, particularly the phrase "be hated by your own mum," which is clearly not meant to be taken literally. The speaker uses this exaggeration to mock or criticise a prior suggestion. However, the model fails to detect this sarcasm and predicts it as non-sarcastic, indicating difficulty in recognising sarcasm that relies on implicit tone rather than explicit linguistic markers.

Text 9 is also labelled as sarcastic (1). The sarcasm in this text is subtle and relies on an implicit contradiction as San Francisco is not typically considered cold, so the statement uses humour to undermine the idea of defining it as such. This requires some degree of contextual understanding. The model misclassifies this as non-sarcastic, suggesting that it struggles to capture sarcasm that depends on external knowledge and indirect cues.

Text 10 is labelled as sarcastic (1). The text presents a narrative describing a chaotic or negative incident but concludes with an overly positive and exaggerated evaluation |("A++ would experience again"), creating a clear contrast between the described event and the final sentiment. This contrast is a key indicator of sarcasm. The model fails to identify this and predicts a non-sarcastic label, likely because it does not effectively track sentiment shifts across longer, multi-sentence inputs.

These four examples were selected because they represent distinct types of misclassification that reveal different weaknesses of the model. Text 3 captures a false positive case where strong emotional language and exaggeration are mistaken for sarcasm, highlighting the model's tendency to over-rely on surface level intensity cues. Contrastingly, Texts 5 and 9 illustrate false negatives where sarcasm is present but expressed implicitly through exaggeration or subtle contradiction, requiring contextual understanding.

Text 10 represents a more complex narrative form of sarcasm, where the sarcastic intent emerges only at the end through a contrast between the described event and the concluding statement. These examples collectively cover a range of linguistic patterns including explicit exaggeration, implicit tone, contextual contradiction, and multi-sentence structure, making them suitable for constructing a few-shot prompt aimed at improving the model's ability to generalise across different forms of sarcasm.

4.3 Results Before and After Prompting

The remaining six misclassified examples from *Table 4.1* were evaluated using the constructed few-shot prompt. Each example was passed into the prompt, which included the four annotated examples along with their correct labels and explanations. *Table 4.2* presents the results before and after applying the few-shot prompt.

Text No.	Text	True Label	Original Predicted Label	Few Shot Predicted Label
1	How insulated are they? I'm over paper thin walls.	0	1	0
2	That's ridiculous. But because people are too embarrassed/shy to say not thanks, it sells at that price and just encourages all prices to be increased. Don't be afraid to walk away if you don't like the price.	0	1	0
4	Yes, if nuclear power plants were free and everyone was happy to have one nearby, they'd threaten all coal power, even some wind and solar. But they're not and they aren't. It's nothing to do with not being "allowed". Literally no energy company is asking for "permission", because they know that nuclear makes no business sense.	0	1	0
6	Yes, "resignation" should not be required - just mark the person as "do not roster". Even if you did have to rehire to cover them, for casuals it doesn't matter, because there is no requirement to take them back after their holiday - that would be a simple decision when they came back of "do I need staff now, or not?" Only reason I can think of is that it would be paperwork to explain why the person was marked "no longer employed" in the payroll and/or rostering computer system. Keeps it neat and tidy?	1	0	1
7	Department of Honesty aka DoH!	1	0	1
8	so you're a genius and a mind reader WOW.	1	0	1

Table 4.2 The results of the few shot prompt tested on the remaining six examples.

4.4 Analysis of Improvement

The few- shot prompt was evaluated using a general-purpose large language model to assess whether prompt-based reasoning could correct the original model's errors [10]. Before applying the few-shot prompt, the TinyLlama LoRA adapter misclassified all six examples, achieving 0/6 correct predictions. This indicates that the model struggled to generalise sarcasm detection beyond its training patterns, particularly for cases involving conversational tone, subtle exaggeration, or implicit intent. After applying the few-shot prompt, all six examples were correctly classified, achieving 6/6 correct predictions. This demonstrates a substantial improvement in performance when the model is provided with explicit examples and reasoning [10].

The improvement can be attributed to several key factors. Firstly, the few-shot prompt provides explicit guidance on how sarcasm should be interpreted, rather than relying solely on learned statistical patterns.

The original LoRA-adapted model appears to rely heavily on surface-level features such as strong emotional language, capitalisation, or sentence structure, which led to systematic errors [11]. For instance, non-sarcastic complaints with exaggerated tone were often misclassified as sarcastic. In contrast, the few-shot prompt demonstrates that intensity of language alone is not sufficient to indicate sarcasm, helping the model recalibrate its decision boundary.

Second, the inclusion of natural language explanations highlights why a statement is sarcastic or not, encouraging the model to focus on deeper semantic cues such as contradiction, intent, and implied meaning [12]. For example, the explanation for sarcastic examples emphasises unrealistic consequences, irony, or contrast between expectation and reality. This allows the model to better recognise sarcasm that is not explicitly marked by obvious keywords.

Third, the selected examples expose the model to multiple forms of sarcasm including exaggeration (e.g. unrealistic outcomes), subtle contradiction (e.g. referencing San Francisco as "cold"), and narrative irony (e.g. a negative event followed by a positive evaluation) [13]. This diversity improves generalisation across different linguistic patterns [10]. As a result, previously misclassified texts such as "Department of Honesty aka DoH!" and "so you're a genius and a mind reader WOW." Were correctly identified as sarcastic, despite relying on implicit tone rather than explicit markers.

Additionally, the prompt helps reduce false positives, where the original model incorrectly labelled literal statements as sarcastic. For example, genuine complaints and reasoned arguments were correctly reclassified as non-sarcastic, indicating that the prompt encourages more careful interpretation of intent rather than reacting to stylistic features alone.

4.5 Conclusion

These results suggest that few-shot prompting enhances performance by shifting the model from pattern-based classification to guided reasoning [10]. While the underlying model parameters remain unchanged, the prompt acts as a form of contextual instruction, enabling the model to apply more nuanced decision-making. However, this improvement was observed on a small subset of examples and the effectiveness of the prompt may vary on larger or more diverse datasets.

Overall, this experiment demonstrates that few-shot prompting can substantially improve sarcasm detection for challenging cases. By providing examples with different types of sarcasm and clear explanations to aid them, the model is better equipped to interpret implicit meaning and reduce common misclassification errors.

5. Deployment And Efficiency

5.1 Deployment Endpoint

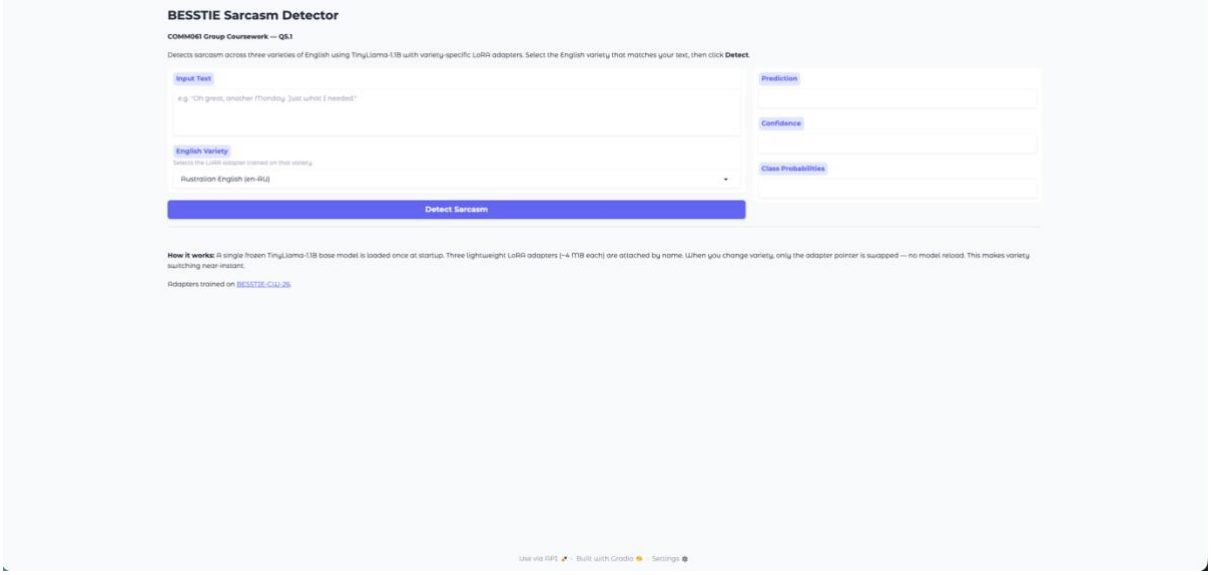


Figure 5.1 Gradio web interface for the BESSTIE Sarcasm Detector showing the text input field and English variety dropdown selector (Australian, Indian, British English)

The deployment is constructed as a Gradio web application that divides the model backend from the front-end interface. The user directly chooses their English variety (Australian, British, or Indian) from a selection menu and enters free-form text. Following the activation of the relevant LoRA adapter, the backend provides a sarcasm prediction, confidence score, and raw class probabilities. The crucial architectural choice is to separate the adapter selection from the base model load. The three variety-specific adapters (en-AU, en-IN, and en-UK) are hosted on HuggingFace and trained on TinyLlama-1.1B using LoRA. The frozen base model is loaded once upon startup, and each of the three adapters is connected by name. There is no model reload when the user chooses a variety; only the active adapter pointer is changed.

Gradio's lightweight, Python-native architecture eliminates the need for independent frontend development, which is why it was chosen as the deployment framework [14]. The variety-specific adapter-switching logic was implemented using Gradio's built-in dropdown component. This deployment's main efficiency improvement is that just the lightweight adapter weights (~4.53 MB per variation) are saved in memory and switched on demand; a single frozen base model (~2.2 GB) is loaded once. In contrast to the approximately 15–30 seconds needed to reload and re-initialize a whole model from disk for every user request, swapping is a pointer reassignment that costs microseconds. This design lowers latency and VRAM load for a service managing multiple concurrent users with varying variety preferences by a factor proportionate to the number of adapters. Three varieties share one base model footprint instead of requiring three full model copies.

5.1.1 Non-Sarcastic vs Sarcastic Examples Prediction Across Varieties

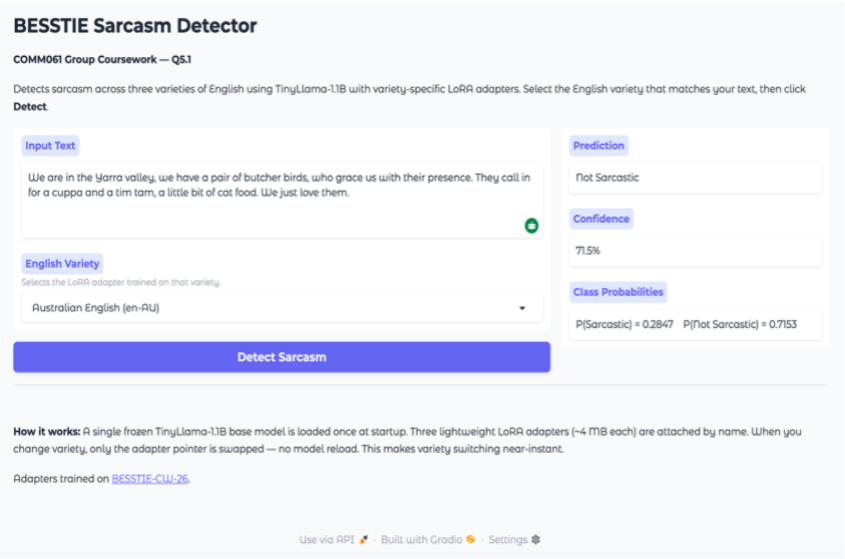


Figure 5.2 en-AU adapter correctly classifies a non-sarcastic Google Places review as Not Sarcastic.

The quantitative assessment is appropriately reflected in the qualitative findings, shown in *Figures 5.2-5.7*. Sarcasm identification is still quite sensitive to training distribution, but non-sarcastic samples are consistently recognized across all dialects. The failures in en-IN and en-UK, where a permanent bias against the majority class is created by extreme class imbalance and culturally distinctive absurdist humor, stand in striking contrast to the successful identification in en-AU. In the end, the deployment acts as an open validation of the model's performance, showing that although the architecture is very effective, data scarcity remains the key obstacle to cross-dialect sarcasm detection rather than deployment limitations [15].

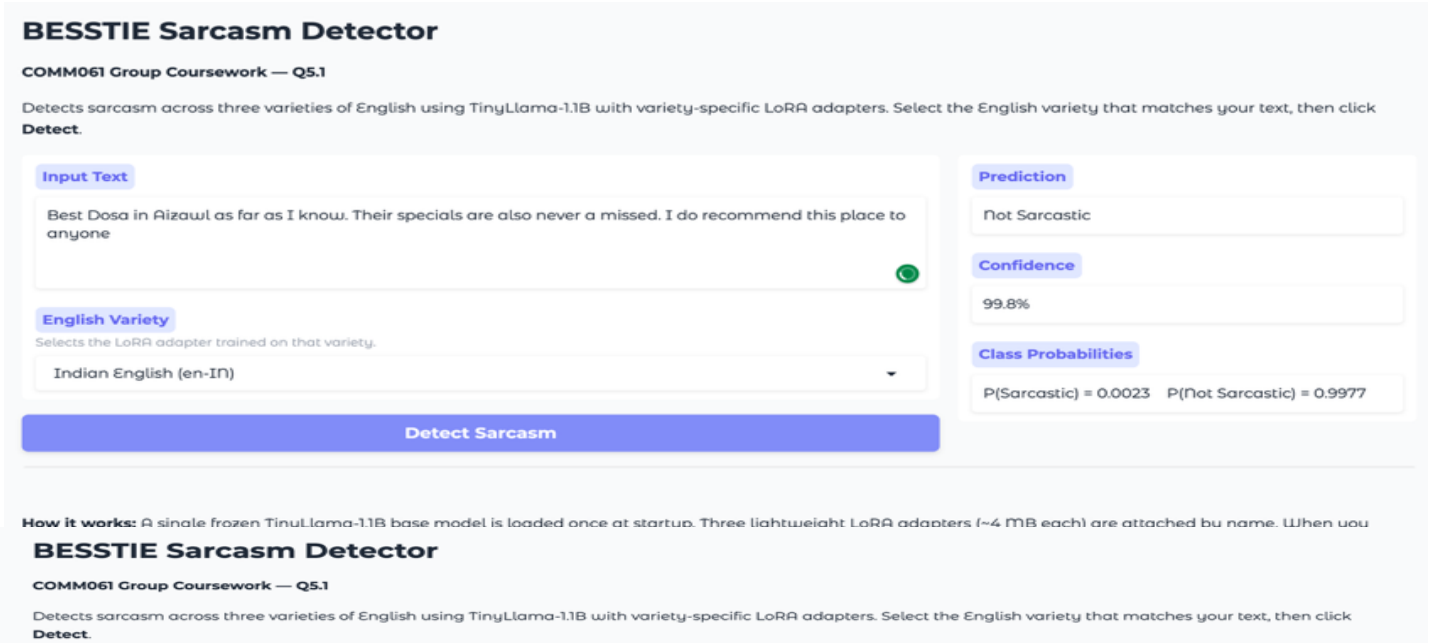


Figure 5.3 en-IN adapter correctly classifies a non-sarcastic Indian English restaurant review with high confidence of 99.8%

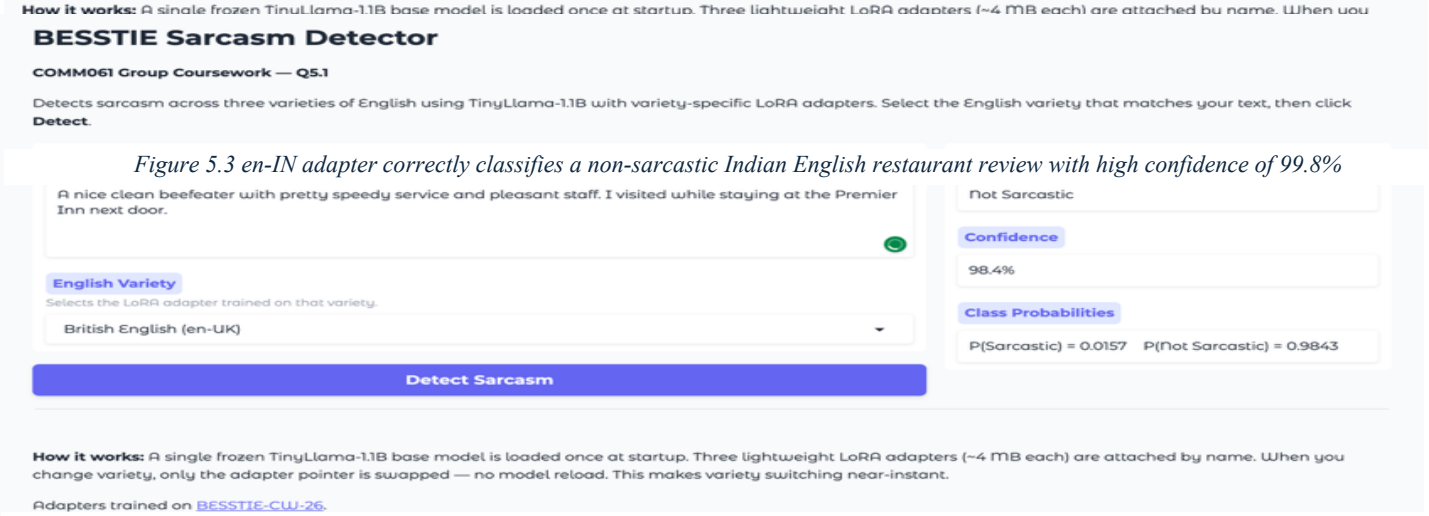


Figure 5.4 en-UK adapter correctly classifies a non-sarcastic British English restaurant review with high confidence of 98.4%

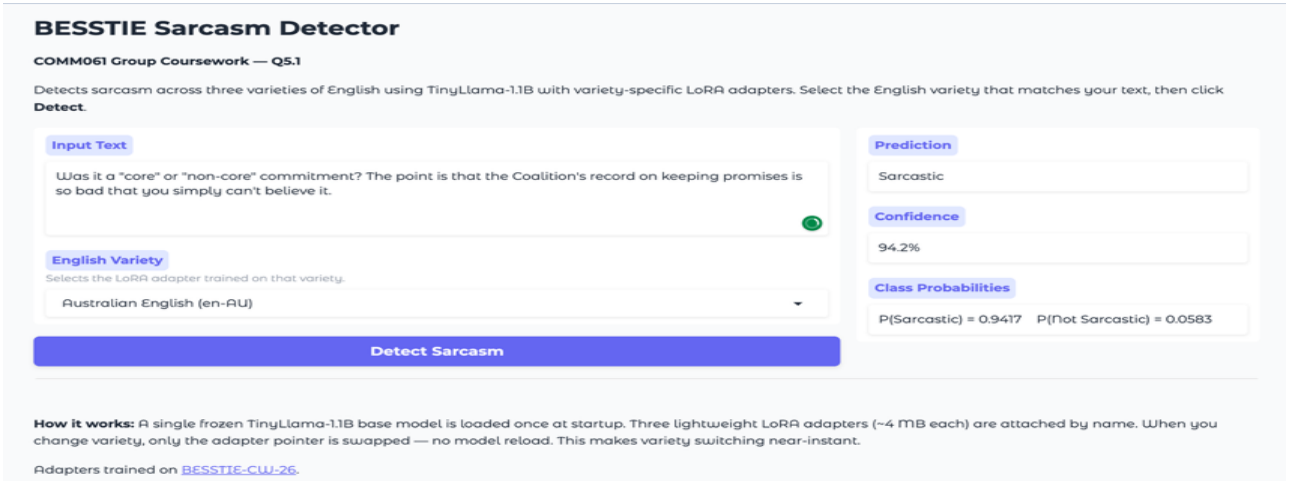


Figure 5.5 en-AU adapter correctly classifies a sarcastic Australian English Reddit comment. High confidence reflects the adapter's stronger sarcasm signal from balanced training data (29.4% sarcastic).

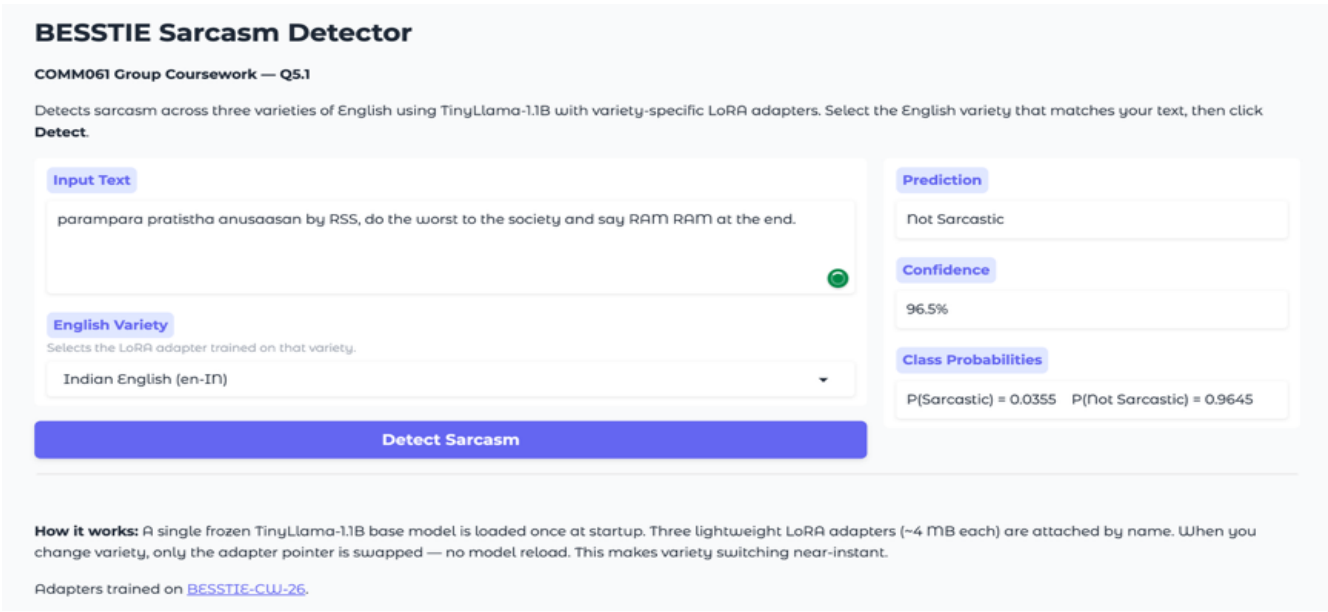


Figure 5.6 en-IN adapter misclassifies a sarcastic code-mixed Hindi-English comment as Not Sarcastic reflectin the adapter's low sarcastic recall due to severe class imbalance in training data.

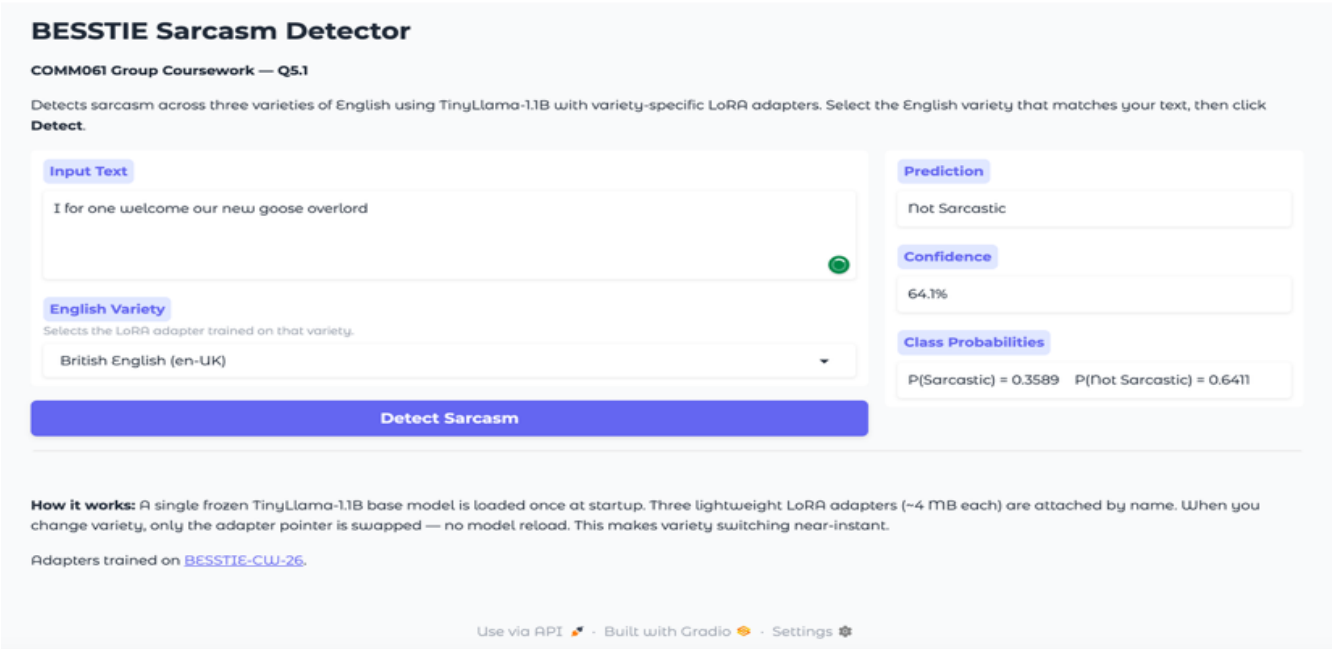


Figure 5.7 en-UK adapter misclassifies a dry British sarcastic comment as Not Sarcastic due to limited training data for sarcasm

5.2 Efficiency

	Average Inference Time (ms)		
Batch Size	**TF-IDF + LR**	**RoBERTa**	**TinyLlama + LoRa**
1	1.0865	24.7159	90.8891
10	0.2537	4.3845	10.741
50	0.0916	2.6306	6.9305
200	0.0709	2.1828	5.9568

Table 5.2 Average inference time for each model across batch size

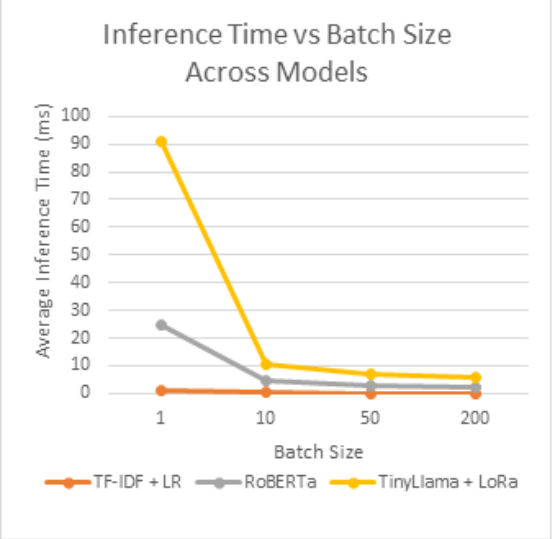


Figure 5.2 Line Graph displaying Average inference time against batch size across different models

To evaluate model efficiency, the average inference time (in milliseconds) was measured across different batch sizes for three models: TF-IDF + Logistic Regression, RoBERTa-base, and TinyLlama-1.1B with LoRA adapters. These results are presented in *Table 5.2* and *Figure 5.2*.

Across all models, inference time decreases as batch size increases [16]. This is expected, as batching allows more efficient use of computational resources by parallelising operations [17]. For example, TinyLlama + Lora decreases from 90.89 ms at batch size 1 to 5.96ms at batch size 200, while RoBERTa decreases from 24.72 ms to 2.18 ms. Even the lightweight TF-IDF baseline benefitted from batching, although its gains are smaller due to its already minimal computational cost.

A clear efficiency gap exists between model types. The TF-IDF + Logistic Regression is by far the fastest, achieving sub-millisecond interference times at largest batch sizes (0.07 ms at batch size 200). This is because it relies on sparse linear operations rather than deep neural computation [18]. Contrastingly, RoBERTa-base introduces significantly higher latency due to its transformer architecture, while the slowest model overall is the TinyLlama + LoRa, reflecting parameter count and token-wise generation [18].

This highlights the key trade-off between model size and response time. Larger models such as RoBERTa and TinyLlama are computationally expensive but better understand context and underlying semantic information [18]. Smaller models like TF-IDF are extremely fast but rely on surface-level features and may struggle with complex linguistic features such as sarcasm [17].

When comparing performance with efficiency, the trade-off becomes clearer. RoBERTa achieves the highest single Macro-F1 score in equivalent settings, while the TF-IDF baseline remains competitive overall for sarcasm detection. TinyLlama + LoRA offers a balance between the two, achieving similar performance to the baseline while enabling variety-specific adaptation, at a higher computation cost.

The choice between speed and performance is depends on the deployment scenario. For real-time or large-scale applications requiring high throughput, a fast model such as TF-IDF may be preferred due to its low latency and scalability [16]. However, for applications where accuracy is more critical, slower transformer-based models may be preferred. In this work, performance would not be fully sacrificed for speed, instead a balanced approach would be favoured. TinyLlama demonstrates that it is possible to achieve competitive performance with parameter-efficient fine-tuning while maintaining reasonable inference times. Therefore, it represents a practical compromise between the simplicity and speed of classical models and the power of large language models.

6. References

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